

Aditya Sugandhi

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SUMMARY

Full-Stack Software Engineer with 5+ years building **scalable support platforms** and **developer tools**. Expert in **TypeScript (React, Next.js)**, **Python**, and **distributed systems** with production experience in **observability**, **debugging infrastructure**, and **API development**. Delivered **monitoring dashboards** and **troubleshooting tools** serving **35K+ devices** with **99.2% reliability**.

TECHNICAL SKILLS

- **Frontend:** React, TypeScript, Next.js, JavaScript, HTML/CSS, Cloudflare Workers, Component Libraries
- **Backend:** Python, FastAPI, Node.js, Java, gRPC, RESTful APIs, GraphQL, WebSocket, Event-Driven Architecture
- **Observability & Debugging:** Grafana, Prometheus, CloudWatch, MLflow, Distributed Tracing, Real-time Monitoring, Alerting
- **Infrastructure:** PostgreSQL, Redis, MongoDB, Apache Kafka, AWS (EKS, Lambda, S3, CloudWatch), Kubernetes, Docker, Linux
- **Developer Tools:** SDK Development, CLI Tools, API Design, Documentation, CI/CD Pipelines, Troubleshooting Automation
- **Practices:** Incident Response, Root Cause Analysis, Cross-functional Collaboration, Agile/Scrum, Security-by-Design

WORK EXPERIENCE

Software Engineer II
Florida State University

Nov 2024 – Present
Tallahassee, FL

- Built **37 RESTful API endpoints** with **real-time WebSocket streaming** serving **35K+ IoT devices**; implemented **comprehensive error handling** and **debugging interfaces** reducing incident resolution time by **40%**.
- Developed **observability infrastructure** with **Grafana dashboards** and **CloudWatch monitoring** for real-time system health tracking; established **automated alerting** for **172+ networks** achieving **99.2% uptime**.
- Architected **developer SDKs** and **tool connectors** enabling cross-system integrations; standardized **event schemas** and **API contracts** improving developer productivity across engineering teams.
- Implemented **fault-tolerant systems** with **Redis caching**, **idempotent retry logic**, and **state management** for distributed workflows; designed **troubleshooting tools** reducing manual investigation time by **40%**.
- Collaborated with cross-functional teams to establish **monitoring best practices** and **incident response procedures**; created **documentation** and **runbooks** for operational excellence.

Machine Learning Researcher
Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024
Tallahassee, FL

- Built **data processing pipelines** handling **1TB+ datasets** using **Python** and **Spark**; implemented **observability dashboards** (Grafana, MLflow) for **quality monitoring** and **latency tracking** across distributed systems.
- Developed **workflow orchestration systems** on **Kubernetes** with **automated retry patterns** and **state management**; achieved **50% pipeline runtime reduction** with **99% job success rate** through optimization.
- Created **Python SDKs** and **debugging tools** for data processing workflows; collaborated with researchers to iterate from experimentation to production deployment with comprehensive logging.

Software Engineer I
Aspire Systems

Oct 2020 – Jul 2022
Chennai, India

- Led **full-stack development** using **Java/Spring** backend and **React/TypeScript** frontend; architected **microservices** processing **500K+ events/day** with **Kafka** event streaming and **99.5% availability**.
- Built **observability pipelines** with **Prometheus/Grafana** for **real-time monitoring** and **alerting**; implemented **distributed tracing** and **debugging interfaces** reducing **MTTR by 60%** through automated root cause analysis.
- Designed **RESTful APIs** and **tool integrations** enabling cross-system data flows; created **internal documentation** and **troubleshooting guides** for support teams.
- Mentored **5 engineers** on **event-driven architecture**, **debugging best practices**, and **security-first coding**; established **code review standards** improving team code quality.

Associate Software Engineer
IMPETUS Technologies

Jun 2019 – Sep 2020
Noida, India

- Developed **Python and TypeScript** backend services for real-time data processing; built **RESTful APIs** handling **100K+ daily requests** with **Redis caching** for low-latency responses.
- Implemented **workflow automation tools** and **monitoring systems** using **Node.js** and **Python**; created internal SDKs for data pipeline orchestration supporting engineering team productivity.
- Built **logging and debugging infrastructure** for data quality monitoring; worked with **PostgreSQL** and **MongoDB** implementing **idempotent operations** and **retry logic** for reliability.

PROJECTS

- **Enterprise Slack Bot Integration** — Deployed **24/7 Slack bot** with **Socket Mode API** for real-time team communication. Configured **OAuth 2.0**, **WebSocket connections**, and **PM2 process management** on **AWS EC2**. Built **monitoring infrastructure** and **troubleshooting CLI tools** achieving **sub-40-second response times**. Tech: **Node.js**, **Next.js**, **AWS**, **WebSocket**, **systemd**.
- **AI Developer CLI & SDK** — Developed **TypeScript-based SDK** for API integration enabling autonomous operations. Built **function-calling interfaces** with **retry logic**, **error handling**, and **idempotency** for enterprise tooling. Created **comprehensive documentation** and **debugging utilities**. Tech: **TypeScript**, **Node.js**, **REST APIs**, **CLI Development**.
- **Full-Stack Developer Platform** — Built **TypeScript/Next.js** frontend and **Python/FastAPI** backend with **real-time workflow orchestration**. Implemented **monitoring dashboards**, **quality scoring**, and **semantic search**. Designed **RESTful APIs** with **comprehensive error handling**. Tech: **TypeScript**, **React**, **Next.js**, **Python**, **FastAPI**, **PostgreSQL**.

EDUCATION

Florida State University
M.S. in Computer Science

Tallahassee, FL

- **Relevant Coursework:** Distributed Systems, Advanced Database Systems, Machine Learning, Data Communications

SRM Institute of Science & Technology
B.Tech. in Computer Science

Chennai, India

PUBLICATIONS & CERTIFICATIONS

- Review Classification & False Feedback Detection (IJAST) | AWS Solutions Architect (Udemy, 2024)